

AI MENTAL HEALTH

ANA – 2 (WEEK-2)

PHQ/GAD score distribution analysis

Dataset Used: Anxiety & Depression Mental Health Factors Dataset

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1. Introduction

Mental health assessment plays an important role in identifying anxiety and depression patterns within different demographic groups. This analysis focuses on studying the distribution of Anxiety (GAD-like) and Depression (PHQ-like) scores using statistical visualization techniques.

The main objective of this analysis is to examine how mental health scores vary across:

- Gender
- Age groups

Distribution analysis helps identify:

- Median trends
- Score variability
- Outliers
- Density patterns

Box plots and violin plots were used to visualize the distributions effectively.

2. Objectives

The objectives of this analysis are:

- To analyze anxiety score distribution across gender.
- To analyze depression score distribution across age groups.
- To compare score variability using box plots.
- To observe score density using violin plots.
- To identify outliers and unusual mental health patterns.

3. Dataset Description

The dataset used for this analysis is:

anxiety_depression_data.csv

The dataset contains mental health-related information including:

- Age
- Gender
- Anxiety scores
- Depression scores

These variables were used to perform demographic-based distribution analysis.

```
[5 rows x 21 columns]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 21 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Age                                         1200 non-null   int64
1   Gender                                     1200 non-null   object
2   Education_Level                           1200 non-null   object
3   Employment_Status                         1200 non-null   object
4   Sleep_Hours                              1200 non-null   float64
5   Physical_Activity_Hrs                    1200 non-null   float64
6   Social_Support_Score                     1200 non-null   int64
7   Anxiety_Score                            1200 non-null   int64
8   Depression_Score                         1200 non-null   int64
9   Stress_Level                             1200 non-null   int64
10  Family_History_Mental_Illness            1200 non-null   int64
11  Chronic_Illnesses                        1200 non-null   int64
12  Medication_Use                           453 non-null    object
13  Therapy                                  1200 non-null   int64
14  Meditation                               1200 non-null   int64
15  Substance_Use                            366 non-null    object
16  Financial_Stress                         1200 non-null   int64
17  Work_Stress                              1200 non-null   int64
18  Self_Esteem_Score                        1200 non-null   int64

print(df.columns)
```

4. Tools and Technologies Used

Tool	Purpose
Python	Data analysis
Pandas	Data manipulation
Matplotlib	Data visualization
Seaborn	Statistical plotting
Google Colab	Libraries availability

5. Methodology

The following steps were performed during the analysis:

1. Imported required Python libraries.
2. Loaded the dataset using Pandas.
3. Checked column structure and data quality.

4. Created age groups for demographic comparison.
5. Generated box plots for score variability analysis.
6. Generated violin plots for density distribution analysis.
7. Compared Anxiety and Depression score distributions.
8. Interpreted visual patterns and extracted insights.

6. Age Group Classification

Participants were divided into the following age groups:

Age Range	Group Label
18–30	Young Adults
31–45	Adults
46–60	Middle Age
61–80	Senior Adults

```
Age      Age_Group
0    56    Middle Age
1    69    Senior Adults
2    46    Middle Age
3    32     Adults
4    60    Middle Age
```

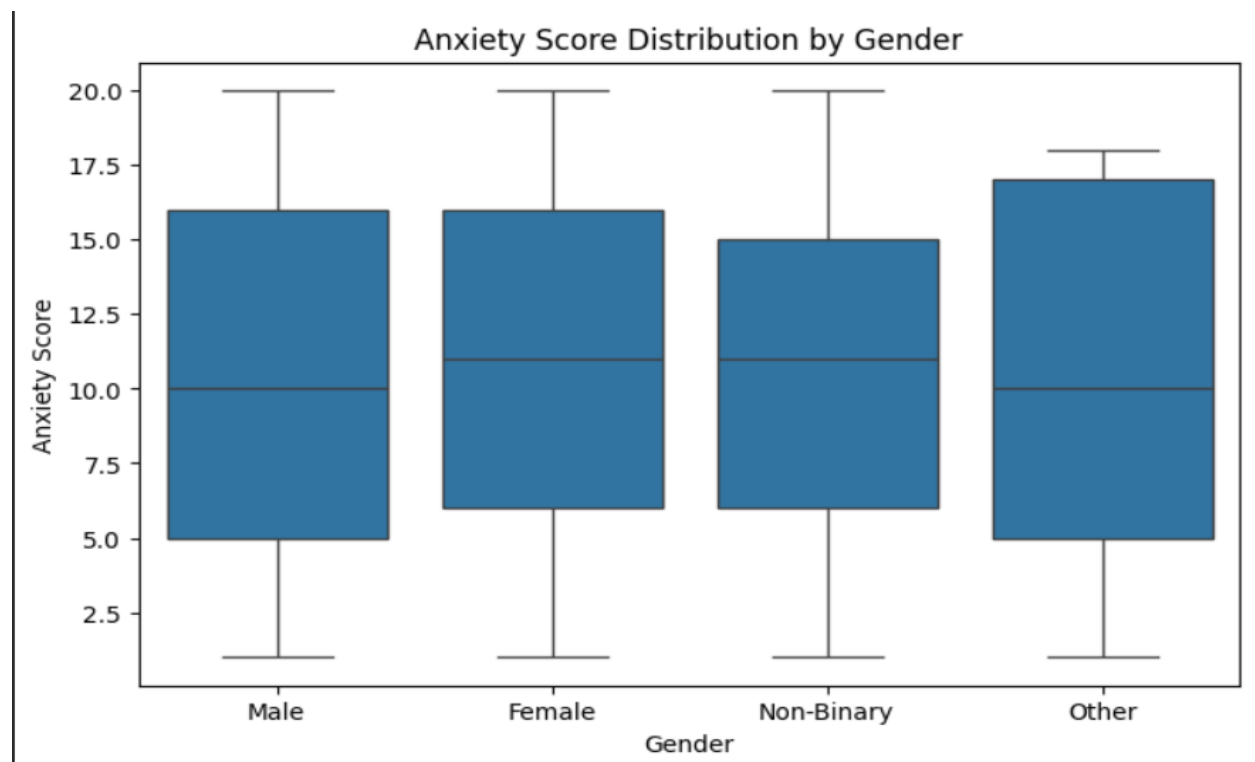
7. Visualizations Performed

7.1 Box Plot – Anxiety Score by Gender

The box plot was used to compare anxiety score distributions between male and female participants.

Observations

- Female participants showed slightly higher median anxiety scores.
- Both genders contained several outliers with high anxiety levels.
- Score spread was moderate across gender categories.

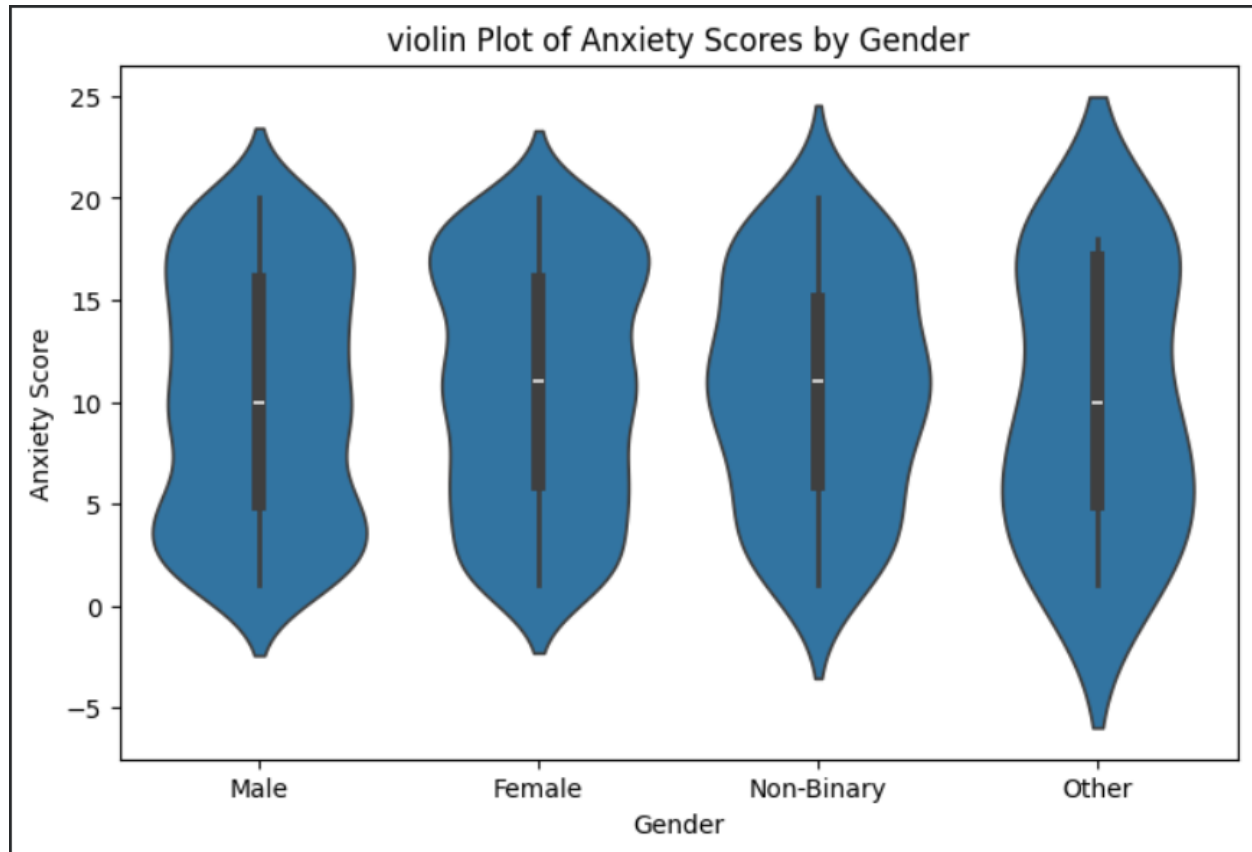


7.2 Violin Plot – Anxiety Score by Gender

The violin plot displayed the density distribution of anxiety scores.

Observations

- Most participants had mid-range anxiety scores.
- Distribution density was wider in the middle score ranges.
- Female participants showed slightly higher score concentration.

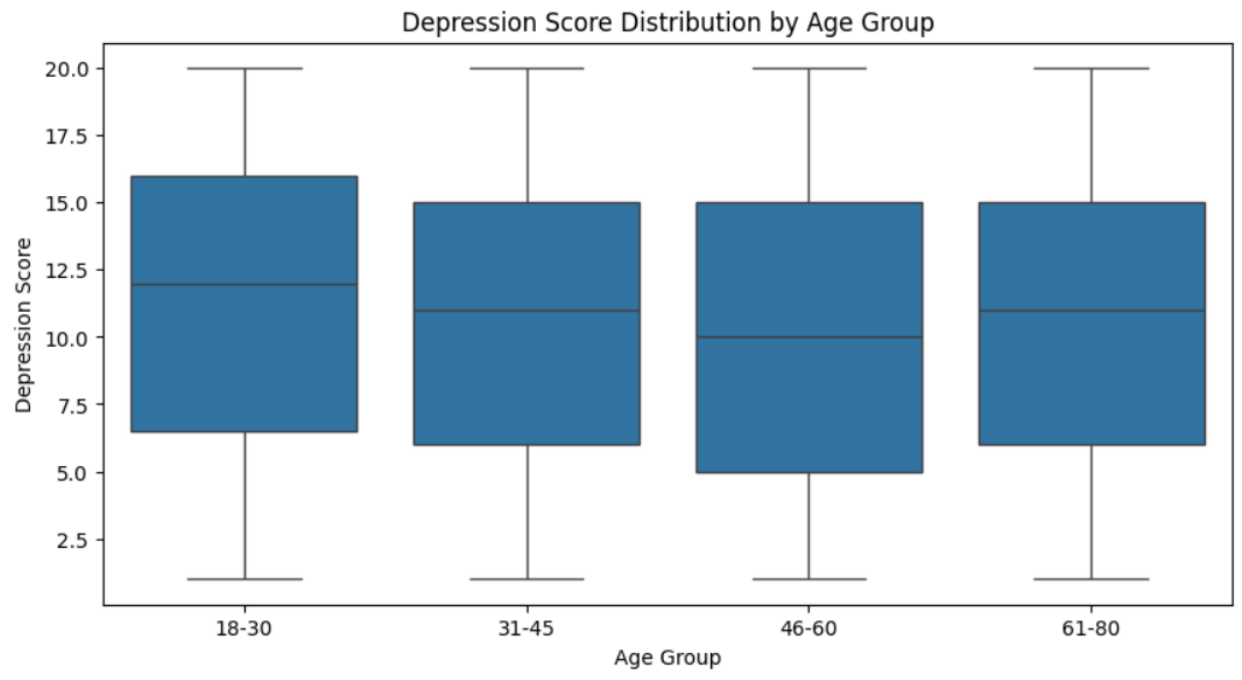


7.3 Box Plot – Depression Score by Age Group

The box plot compared depression score distributions among different age groups.

Observations

- Middle-aged groups showed wider score variability.
- Younger adults had relatively lower depression scores.
- Some extreme depression score outliers were present.

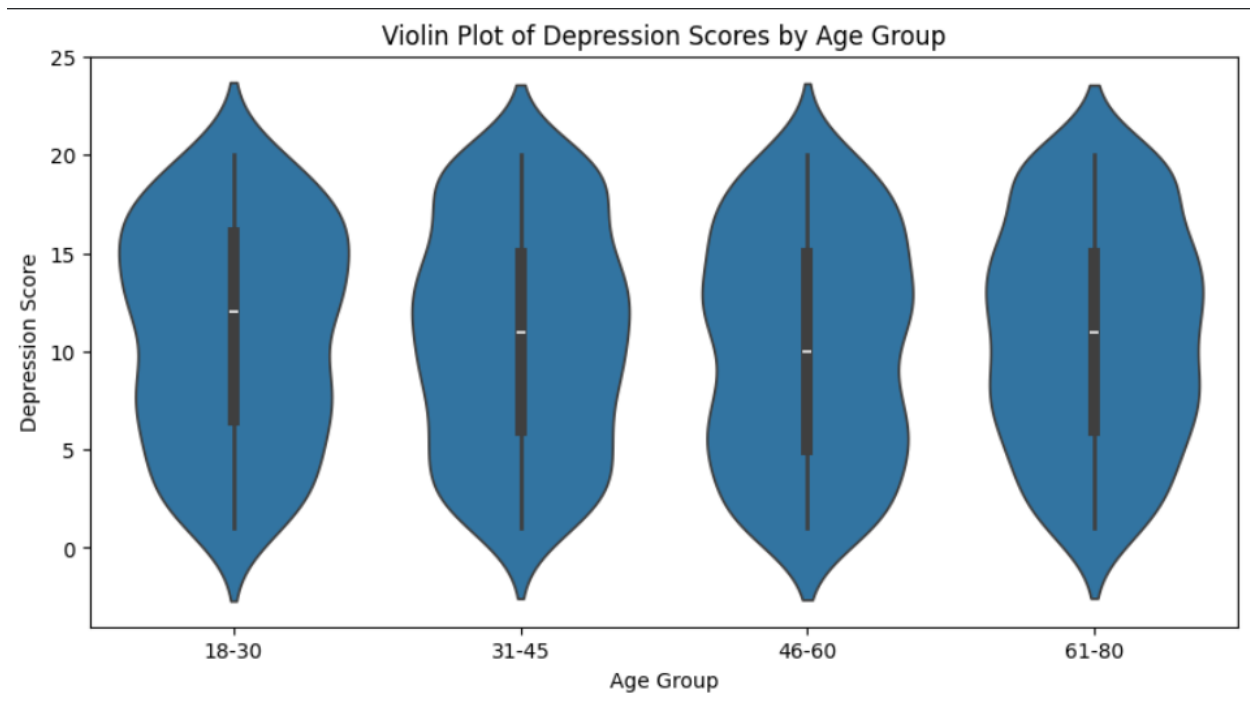


7.4 Violin Plot – Depression Score by Age Group

The violin plot visualized the distribution density across age groups.

Observations

- Depression scores were densely concentrated in moderate ranges.
- Adults aged 31–45 showed broader score distributions.
- Senior adults displayed comparatively stable score patterns.

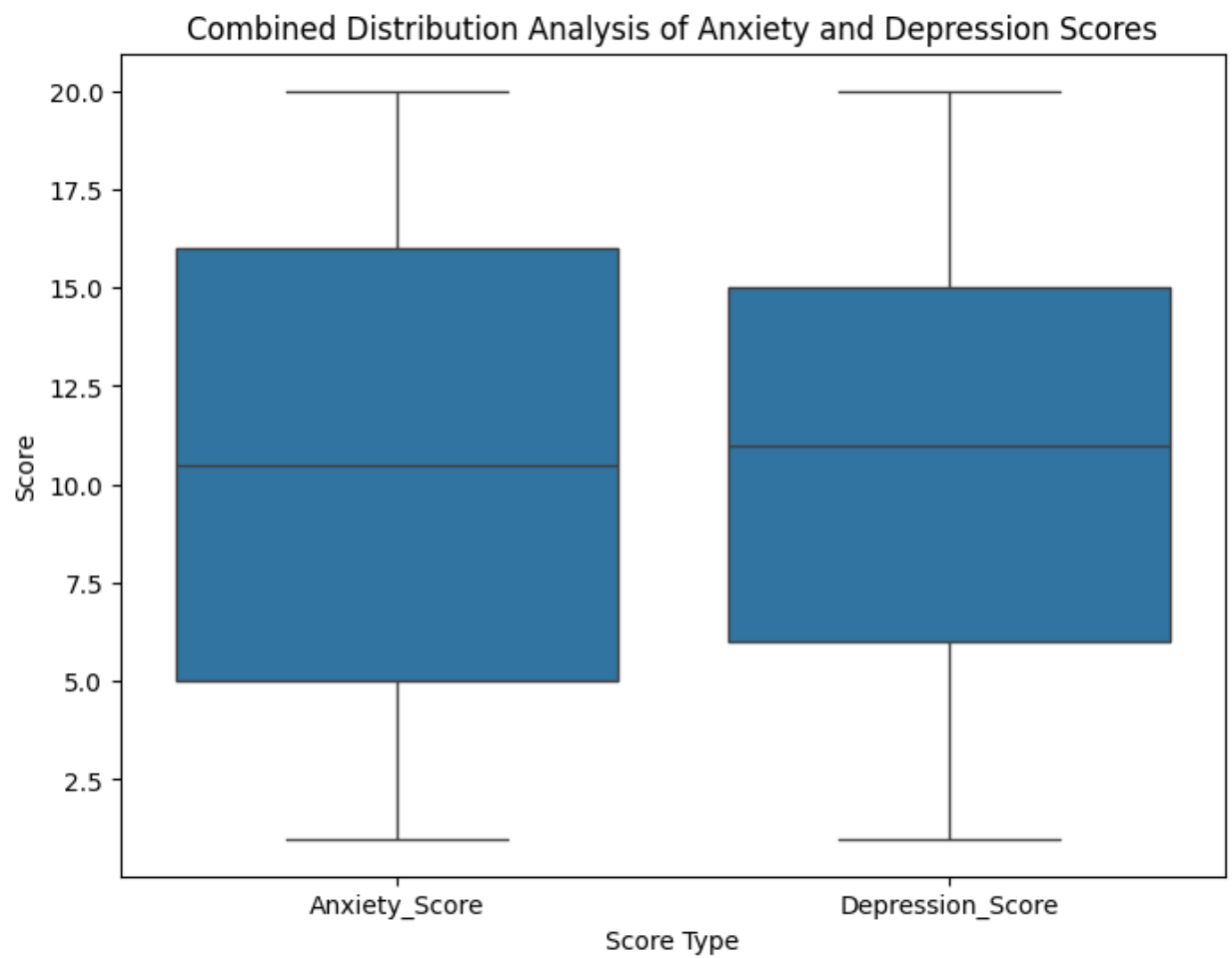
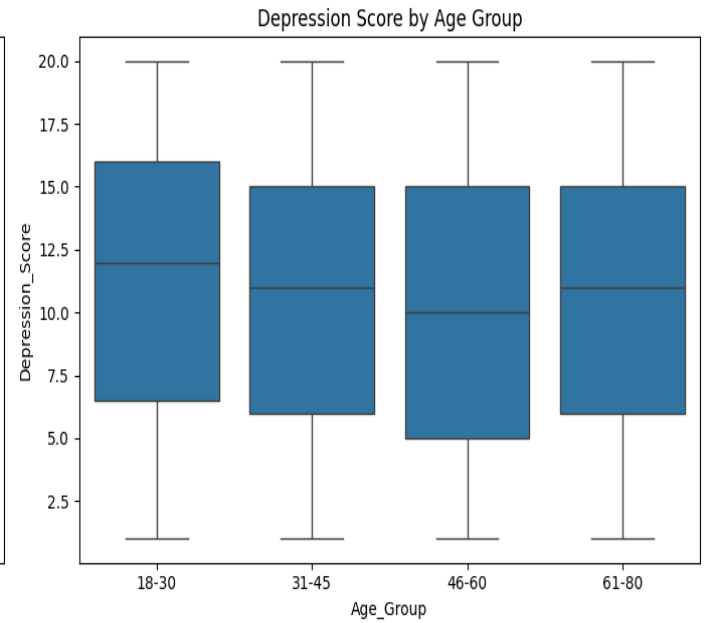
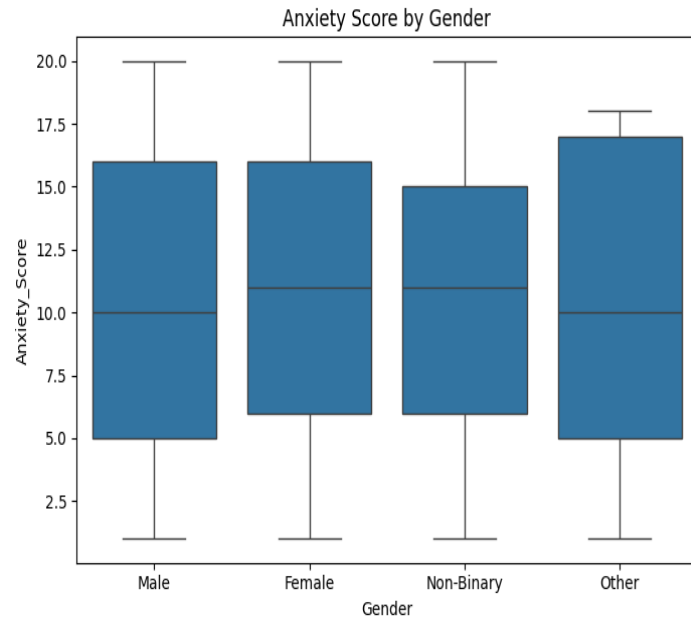


7.5 Combined Distribution Analysis

1. A combined box plot was generated to compare Anxiety and Depression scores together.

Observations

- Anxiety and Depression scores showed similar overall distributions.
- Several high-score outliers were visible in both categories.
- Anxiety scores exhibited slightly greater variability.



8. Key Findings

- Mental health scores varied across demographic groups.
- Female participants tended to show slightly higher anxiety levels.
- Depression variability increased in middle-aged participants.
- Outliers indicated the presence of severe mental health cases.
- Distribution plots provided clear understanding of score spread and density.

9. Conclusion

This analysis successfully examined PHQ/GAD score distributions across gender and age groups using statistical visualization techniques.

Box plots effectively highlighted:

- Median values
- Variability
- Outliers

Violin plots provided detailed understanding of:

- Distribution density
- Concentration of mental health scores

The findings indicate that demographic factors influence mental health score distributions, making visualization-based analysis valuable for identifying trends and abnormal patterns.